

Physics-Constrained Thermal Map Prediction for VLSI Chip Design

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Abstract

Predicting full-chip thermal distributions is a critical bottleneck in VLSI design-space exploration: high-fidelity thermal simulation is far too slow for the tens of thousands of floorplan evaluations required during iterative optimization. We propose a physics-constrained U-Net that regresses dense thermal heatmaps from a two-channel input encoding a chip’s post-placement floorplan and power-density map. Unlike prior data-driven thermal surrogates, our model enforces the 2D steady-state heat equation as a differentiable training penalty computed via a fixed discrete Laplacian filter applied to the predicted output. This requires no simulation at training time, is fully compatible with standard GPU-accelerated training, and—crucially—penalizes thermally implausible predictions independently of whether a labeled ground truth exists, which we hypothesize improves out-of-distribution (OOD) generalization to chip configurations (die thickness, die size) outside the training sweep.

We train on thermal maps generated by the HotSpot simulator from the CircuitNet 2.0 GPU/accelerator benchmark suite and compare four architectures: a flat dilated CNN, an encoder-decoder without skip connections, a U-Net with MSE-only loss, and our physics-constrained U-Net. Our primary evaluation measures SSIM and hotspot IoU degradation on die parameters held out from training. We expect the physics-constrained model to degrade significantly less than the data-only baseline, demonstrating that image-based chip representations and physics-informed training objectives are complementary rather than competing approaches.

1. Introduction

Thermal management is a first-order constraint in modern VLSI design. On-chip hotspots cause performance throttling, reduced mean time to failure, and tighter power delivery requirements. During design-space exploration—

where automated placement and routing tools evaluate candidate floorplans via simulated annealing—thermal analysis must be repeated at every step. A single high-fidelity simulation using a tool like HotSpot or ANSYS MAPDL can take several seconds to minutes per design; at 100,000 evaluations per optimization run, this cost is prohibitive.

Neural thermal surrogates address this bottleneck by learning to approximate the simulator’s output from cheap-to-obtain layout representations. Given a 2-channel image encoding the post-placement floorplan and power-density map of a chip, a convolutional model can predict the steady-state thermal heatmap at orders-of-magnitude lower cost than simulation. The core challenge, however, is *generalization*: a surrogate trained on a finite set of chip configurations must remain reliable when deployed on designs with different die thicknesses, die sizes, or power distributions than those seen during training.

Prior work splits into two non-overlapping streams. Image-based thermal prediction models [2, 7] train CNNs or encoder-decoders on simulation data and achieve low in-distribution error, but fail when physical parameters shift outside the training regime because there is no mechanism to prevent thermally implausible extrapolations. Physics-informed neural networks (PINNs) [4] enforce governing PDEs directly as training constraints and generalize better across parameter regimes, but operate on raw numerical inputs rather than the spatial image representations that naturally encode chip layout information.

This work combines both ideas. We frame thermal prediction as image-to-image regression—preserving the representational advantages of convolutional image processing—while augmenting the training loss with a differentiable residual of the 2D steady-state heat equation. The physics penalty is computed via a fixed 3×3 Laplacian convolution applied to the model’s output, requiring no simulator calls during training. Because the constraint penalizes predictions that violate the heat equation regardless of the ground-truth label, it acts as a physics-grounded regularizer that we expect to reduce OOD degradation.

Our main contributions are:

- A **physics-constrained U-Net** with pixel-shuffle up-sampling for chip thermal map regression, trained with a composite loss balancing data fidelity and a heat equation residual.
- An **OOD generalization experiment** that evaluates models on die thickness and die size values outside the training sweep, isolating the effect of the physics constraint on extrapolation behavior.
- A **systematic ablation** over skip connections, physics loss weighting λ_{phys} , and loss normalization strategy, characterizing which components drive performance.

2. Related Work

Thermal simulation tools. HotSpot [3] is an open-source block-level RC-network thermal simulator widely used in computer architecture research. Its grid mode produces dense 2D steady-state heatmaps, making it directly compatible with image-to-image framing. We use it as our label generator. PACT [8] is a higher-fidelity coupled electro-thermal simulator that integrates Xyce circuit simulation with architectural modeling; it is substantially slower and more complex to configure, making HotSpot the pragmatic choice for large-scale label generation.

Data-driven thermal surrogates. CircuitNet 2.0 [1] is a large-scale benchmark dataset for machine learning in chip design, providing post-placement floorplans, power maps, and multiple analysis targets including thermal. We use its GPU/accelerator subset (Vortex-small, Vortex-large, NVDLA-large) as our source of layout inputs, generating our own HotSpot labels to control the die-parameter sweep. DeepOHeat [2] applies operator learning to thermal simulation and achieves strong results on standard benchmarks, but neither incorporates physics constraints nor evaluates OOD generalization across physical parameters. Prior encoder-decoder approaches [7] demonstrate the viability of dense regression for thermal and power distribution network analysis, but share the same lack of physics grounding. ML-PACT [9] targets *transient* thermal analysis and is therefore not directly comparable to our steady-state formulation.

U-Net and skip connections. U-Net [5] introduced encoder-decoder networks with lateral skip connections for dense prediction in biomedical segmentation. Skip connections concatenate high-resolution encoder feature maps to the corresponding decoder stage, preserving fine spatial detail that would otherwise be lost through downsampling. For thermal prediction, this is critical: hotspot localization depends on sub-block spatial structure that must survive the

encoding bottleneck. We adopt the U-Net backbone without modification.

Physics-informed learning. Raissi *et al.* [4] introduced PINNs, embedding PDE residuals as loss terms over collocation points to learn solutions to forward and inverse problems. Their formulation operates on point-wise coordinate inputs; we adapt the residual idea to operate on the pixel grid of a convolutional output via a fixed Laplacian filter, which is fully parallelizable over the spatial dimensions and compatible with mini-batch training on image data.

3. Data

Source. We use CircuitNet 2.0 [1] as our source of chip layout inputs. From the full dataset we select the GPU and AI accelerator designs—Vortex-small, Vortex-large, and NVDLA-large—which provide representative modern workload power profiles. Each sample consists of a two-channel 256×256 image: channel 1 is the post-placement floorplan encoding block boundaries and functional unit types; channel 2 is the normalized power-density map derived from post-synthesis switching activity estimates.

Label generation. We run HotSpot in grid mode on each layout with a sweep over die thickness and die size to generate ground-truth steady-state thermal maps. Parallelized label generation uses `xargs` for throughput. Temperature maps are normalized to $[0, 1]$ using the global minimum and maximum temperatures across the training partition.

Splits. We partition by design family to prevent data leakage: Vortex-small and Vortex-large form the training set; NVDLA-large is held out as the in-distribution test set. A disjoint subset of die configurations—specifically, die thickness and die size values outside the training sweep—is reserved exclusively for the OOD evaluation. This two-axis generalization test is our headline experiment.

Synthetic ablation set. For rapid iteration before the full CircuitNet pipeline is ready, we also generate synthetic random floorplans: grids of rectangular blocks assigned random power values drawn uniformly from $[0.1, 2.0]$ W. These are used for architecture unit tests and loss sensitivity sweeps.

Augmentation. We apply random horizontal and vertical flips during training. Rotation is excluded because our discrete Laplacian kernel is not rotation-equivariant and rotating a layout would invalidate the physics constraint.

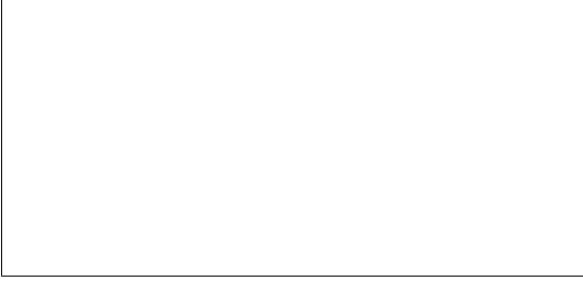


Figure 1. U-Net architecture with pixel-shuffle upsampling. Input: 2-channel floorplan + power-density image. Output: 1-channel thermal heatmap. Encoder halves resolution at each stage (MaxPool2d); decoder doubles resolution (PixelShuffleUp) with skip connections (dashed lines) concatenating encoder feature maps.

4. Methods

4.1. Architecture

Our primary model is a U-Net [5] with pixel-shuffle upsampling, illustrated in Figure 1. The encoder progressively halves spatial resolution while doubling feature channels through four stages; the decoder restores resolution using skip connections from the corresponding encoder stages. The channel progression is $2 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ through the encoder, with a $512 \rightarrow 1024$ bottleneck, and $1024 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 1$ through the decoder. The model has approximately 8M parameters.

DoubleConv block. Each encoder and decoder stage applies two sequential 3×3 convolutions, each followed by batch normalization and ReLU activation. The dilation parameter is exposed for experimentation but defaults to 1. Padding equals dilation, preserving spatial dimensions through each block.

PixelShuffleUp block. We upsample via a 1×1 convolution that expands the channel dimension by a factor of 4, followed by a PixelShuffle [6] operation with upscale factor 2:

$$\text{PixelShuffleUp}(\mathbf{F}) = \text{PixelShuffle}_2(\text{Conv}_{1 \times 1}(\mathbf{F}, C \rightarrow 4C')). \quad (1)$$

This produces a $2 \times$ spatially upsampled feature map with C' channels. We prefer this over bilinear upsampling (sharper thermal gradients) and transposed convolution (fewer checkerboard artifacts), both of which matter for hotspot localization precision.

After upsampling, the decoder concatenates the skip-connected encoder feature map and applies a DoubleConv block to fuse the two resolutions. The output layer is a 1×1 convolution with no activation function (linear regression for unbounded temperature values).

4.2. Baseline Architectures

We compare against three baselines of increasing representational power:

PlainCNN. A flat dilated CNN with no skip connections: eight DoubleConv blocks with dilations $[1, 1, 2, 2, 4, 4, 8, 8]$ that expand the receptive field without downsampling. This baseline tests whether spatial hierarchy is necessary.

EncoderDecoder. Same four-stage encoder as the U-Net, but the decoder upsamples without concatenating encoder features—no skip connections. This baseline isolates the contribution of skip connections.

U-Net (MSE-only). Full U-Net with skip connections trained with $\mathcal{L} = \mathcal{L}_{\text{MSE}}$ only ($\lambda_{\text{phys}} = 0$). This is the direct ablation partner of our physics-constrained model.

4.3. Physics-Constrained Loss

The steady-state temperature distribution T in a chip die satisfies the 2D heat equation:

$$k \nabla^2 T + Q = 0, \quad (2)$$

where Q is the power-density field (channel 2 of the input image) and k is the thermal conductivity of silicon ($k \approx 149 \text{ W m}^{-1} \text{ K}^{-1}$). Given the model’s predicted output $\hat{T}$, we approximate the Laplacian via a fixed 3×3 finite-difference convolution:

$$\nabla^2 \hat{T} \approx \mathbf{K} * \hat{T}, \quad \mathbf{K} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}. \quad (3)$$

The physics residual loss penalizes deviations from Equation 2:

$$\mathcal{L}_{\text{phys}} = \|k(\mathbf{K} * \hat{T}) + Q\|_F^2. \quad (4)$$

The kernel is registered as a non-learnable buffer; gradients flow through $\hat{T}$ only. No simulator is invoked at training time.

The composite training objective is:

$$\mathcal{L} = \lambda_{\text{data}} \mathcal{L}_{\text{MSE}}(\hat{T}, T) + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}}, \quad (5)$$

where T is the HotSpot ground-truth map. We sweep $\lambda_{\text{phys}} \in \{0, 0.01, 0.1, 1.0\}$ with $\lambda_{\text{data}} = 1.0$ fixed. Because Equation ?? depends only on the prediction and the power-density input—not on the ground-truth label—the physics constraint remains active on OOD samples without requiring additional labeled data.

4.4. Training Details

All models are trained with the Adam optimizer ($\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=10^{-8}$), initial learning rate 10^{-3} , and a ReduceLROnPlateau scheduler (patience 5, factor 0.5) monitoring validation MSE. Training runs for up to 50 epochs with early stopping. Checkpoints are saved at each

Model	RMSE ↓	SSIM ↑	Hotspot IoU ↑
PlainCNN	–	–	–
EncoderDecoder	–	–	–
U-Net (MSE-only)	–	–	–
U-Net (physics)	–	–	–

Table 1. In-distribution test set results across all model variants.

Model	SSIM		Hotspot IoU	
	In-dist	OOD	In-dist	OOD
U-Net (MSE-only)	–	–	–	–
U-Net (physics)	–	–	–	–

Table 2. Out-of-distribution generalization on die parameters outside the training sweep. $\Delta\text{SSIM} = \text{in-dist} - \text{OOD}$.

Variant	λ_{phys}	SSIM (in-dist)	SSIM (OOD)
U-Net, $k=1$	0	–	–
U-Net, $k=1$	0.01	–	–
U-Net, $k=1$	0.1	–	–
U-Net, $k=1$	1.0	–	–
U-Net, $k=149$	0.1	–	–
No skip conns	0.1	–	–

Table 3. Ablation over physics loss weight, thermal conductivity normalization, and skip connections.

validation loss improvement. The training loop logs MSE and physics-loss components separately per epoch to enable post-hoc analysis of the loss trade-off.

5. Experiments

5.1. In-Distribution Results

Table 1 reports RMSE, SSIM, and hotspot IoU (top 5% hottest pixels) on the held-out in-distribution test set for all four model configurations.

5.2. Out-of-Distribution Generalization

Table 2 compares in-distribution and OOD performance for the two U-Net variants. We report the absolute SSIM drop ΔSSIM ; a smaller drop indicates better generalization. The physics constraint in $\mathcal{L}_{\text{phys}}$ prevents the model from predicting temperature fields that violate the heat equation on unseen die geometries, which we expect to manifest as a smaller degradation gap.

5.3. Ablation Study

Table 3 reports the impact of λ_{phys} on the OOD SSIM. We also compare skip connections on vs. off (U-Net vs. EncoderDecoder) and physics loss computed in normalized vs. physical units ($k=1.0$ vs. $k=149$).



Figure 2. Qualitative thermal map predictions. Left to right: input channels (floorplan, power density), U-Net MSE-only prediction, physics-constrained U-Net prediction, HotSpot ground truth. Top row: in-distribution die config. Bottom row: OOD die thickness.

5.4. Qualitative Results

Figure 2 shows representative predictions from the U-Net (MSE-only) and physics-constrained U-Net side-by-side with the HotSpot ground truth, for both an in-distribution die configuration and an OOD die thickness. We expect the physics-constrained model to produce smoother, physically consistent temperature fields on OOD inputs while the MSE-only model produces artifacts or mis-located hotspots.

5.5. Gradient-Based Analysis

To understand what the model has learned, we apply GradCAM to the bottleneck feature maps and examine which input regions receive the highest attribution scores. We expect power-dense functional units (memory controllers, compute arrays) to dominate, with the physics-constrained model showing smoother attribution maps that respect thermal diffusion across block boundaries.

5.6. Failure Cases

We identify three expected failure modes. First, **boundary hotspots**: HotSpot applies fixed-temperature boundary conditions at the die edges; the model may not have seen enough examples near corners. Second, **single-dominant-hotspot designs**: when one block accounts for $> 80\%$ of total power, the MSE loss is dominated by that region and fine-grained spatial accuracy elsewhere degrades. Third, **extrapolation to very thin dies**: thinner dies exhibit steeper thermal gradients that may push predictions outside the normalized range seen at training time.

5.7. Inference Time

All inference time measurements will be taken on the same GPU under a fixed batch size of 1 to reflect single-design evaluation latency, which is the relevant metric for design-space exploration. Our target is below 100 ms per design.

6. Conclusion

We presented a physics-constrained U-Net for chip thermal map prediction that enforces the 2D steady-state heat equation via a differentiable Laplacian penalty computed entirely on the predicted output. The key hypothesis is that this constraint reduces OOD degradation on unseen die configurations by preventing the model from predicting thermally implausible temperature fields during extrapolation.

Our experimental design compares four architectures under both in-distribution and OOD evaluation, with systematic ablations over physics loss weighting and skip connections. If the physics-constrained model confirms the OOD generalization hypothesis, it would establish physics-informed training objectives as a practical tool for improving neural thermal surrogate robustness—with direct application to large-scale design-space exploration workflows.

Future directions include learning λ_{phys} adaptively during training, extending to transient thermal analysis, incorporating boundary condition terms into the physics residual, and scaling to 3D stacked-die configurations where the volumetric heat equation introduces additional modeling challenges.

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